

Technology Stack

Date	03 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML, CSS, Bootstrap, JavaScript</i>	<i>Used to develop a responsive and user-friendly web interface that allows users to enter applicant details and view credit card approval predictions.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, Flask</i>	<i>Flask provides a lightweight web framework for handling user requests, processing input data, loading the trained machine learning model, and displaying prediction results.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset (Application Record & Credit Record)</i>	<i>Historical applicant data stored in CSV files is used for data preprocessing, model training, validation, and testing. (If you used a database like MySQL or SQLite, replace this accordingly)</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>IBM Watson Machine Learning, IBM Cloud</i>	<i>Used to deploy the trained machine learning model on the cloud, enabling scalable, real-time prediction services that can be accessed through the web application.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Git enables version control, while GitHub facilitates collaborative development, code management, and project documentation among team members.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>IBM Watson Machine Learning API, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn</i>	<i>IBM Watson Machine Learning API enables cloud deployment and prediction. Scikit-learn and XGBoost are used to build and evaluate machine learning models. Pandas and NumPy support data preprocessing, while Matplotlib and Seaborn are used for exploratory data analysis and visualization.</i>